

MD. RAKIB HOSSAIN

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Computer Science and Engineering student specialising in machine learning and data science. Builds research-grade ML systems end to end: calibrated and uncertainty-aware models, computer-vision detection pipelines, custom datasets built from scratch, and honest evaluation that reports where a model fails as clearly as where it works. Comfortable shipping the result as a working application, and experienced in leading a technical team from concept to delivery.

TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript/TypeScript

Machine Learning: PyTorch, TensorFlow, scikit-learn, YOLO, OpenCV, SHAP

Data: pandas, NumPy, Matplotlib, Microsoft 365 (Excel, Word, PowerPoint)

Software and Tools: Git and GitHub, Jupyter, Google Colab, Claude Code, Antigravity, Roboflow, Next.js, NestJS, PostgreSQL

PUBLICATIONS

- *DhakaNight: A Benchmark for Low-Light Object Detection in Dense Urban Night Traffic*. Accepted, 4th International Conference on Computing Advancements (ICCA 2026), Dhaka. Proceedings submitted to the ACM Digital Library; prior editions indexed in Scopus.
- *PulseStone: A Tangible Single-Action Anxiety Companion*. Accepted, 4th International Conference on Computing Advancements (ICCA 2026), Dhaka.

PROJECTS

RideGuard - Calibrated and Auditable ML Risk Framework | Python, CatBoost, scikit-learn, SHAP

- Crash-risk model combining a calibrated CatBoost scorer with a DBSCAN and KDE spatial prior: 0.973 macro-F1, calibration error cut 5.6x (ECE 0.043 to 0.008), conformal coverage 0.895 against a 0.90 target.
- Outperformed MLP, TabNet and Keras baselines; added ADWIN drift detection and SHAP explanations.
- Self-audit showed the model was fitting a near-deterministic label rather than real pre-crash signal, reported as the central finding. Deployed as a live web app with risk map and route-level alerts.

[Research](#) | [App Code](#) | [Live](#)

DhakaNight - Low-Light Dataset Creation and Detection Benchmarking | Python, PyTorch, YOLO, OpenCV, Roboflow

- Built and annotated a 2,300-image low-light street dataset in Roboflow to cover a gap in existing detection data.
- Benchmarked CLAHE, Gamma, Zero-DCE and RetinexFormer against a YOLOv8 baseline (0.587 mAP@50): no enhancement method improved detection, RetinexFormer cost 5.4 points, and simply scaling the detector beat every preprocessing pipeline.

[Code](#)

Drikon - Full-Stack E-Commerce Platform | Next.js, NestJS, PostgreSQL, Prisma, Redis

- Monorepo storefront and REST API with two-factor authentication, rotating JWTs, argon2id hashing and role-based access control, plus an admin panel for product and order management.

[Live](#) | [Code](#)

PulseStone - Embedded Device and Controlled User Study | ESP32-C3, Arduino C++, Python

- Built a handheld breathing-guide device on an Arduino C++ finite-state machine with synchronized LED, haptic and display feedback.
- 15-participant study over 150 trials: 14.5x faster initiation, 7.49 s to 0.52 s ($t(14) = 70.5$, $p < 0.001$, Cohen's $d = 18.2$).

[Code](#)

LEADERSHIP AND ACTIVITIES

Founder and Team Lead, "Khepa Chakka" Robotics Team

2023 - Present

- Founded and lead a competitive robotics team, designing, building and programming four classes of robot (line-following, soccer, race, and battle), covering sensor integration, motor control, and real-time embedded logic.
- Coordinate members across mechanical design, electronics, and software against competition deadlines; led the team to six national and international podium finishes, including selection for Techfest, IIT Bombay.

AWARDS AND HONORS

- Winner, Cozmo Clench, Techfest IIT Bombay (Bangladesh Zonal), 2024 - one of Asia's largest science and technology festivals.
- Champion, Robot Soccer, AIUB CS Fest 2024; 1st Runner-up, Robot Soccer, AIUB Robotic Crew (ARC).
- 1st Runner-up (Nano Battle Bot) and 2nd Runner-up (Robo Race), National Robotics Championship 2025; 1st Runner-up, Robo Race, Technoxian Bangladesh National Round 2024.

EDUCATION

B.Sc. in Computer Science and Engineering, American International University-Bangladesh (AIUB)

Expected 2026

CGPA: 3.28 / 4.00

HSC, Dhaka Imperial College, 2021 | SSC, Dhaka Collegiate School, 2019 | **Languages:** Bangla (Native), English